Submission A solution to Problem 5

Uniqueness of the invariant measure

Problem statement and interpretation

Put $E = C_0([0,1])$, $H = L^2(0,1)$, and let S_t be the Dirichlet heat semigroup generated by $\frac{1}{2}\Delta$. I use the Dirichlet version of the Athreya–Butkovsky–Lê–Mytnik approximation assumption in the following form. For each deterministic initial condition $x \in E$, if $b_n = G_{1/n}$, then the regularized equation has the mild solution

$$u_t^n = S_t x + V_t + K_t^n, \qquad K_t^n = \int_0^t S_{t-r} b_n(u_r^n)\,dr, \qquad V_t = \int_0^t S_{t-r}\,dW_r,$$

and, for every $t > 0$ and every $\Phi \in C_b(E)$,

$$P_t^n \Phi(x) := \mathbb{E}\Phi(u_t^n) \longrightarrow P_t \Phi(x). \tag{1}$$

Thus I interpret the stated weak convergence of $u^n(t)$ to $u(t)$ as holding for all deterministic starting points; this is the form needed for a statement about invariant measures of the whole Markov semigroup.

Lemma 1 (Uniform estimate on the regularized drift). *For each $T < \infty$ there is $C_T < \infty$, independent of n and $x \in E$, such that*

$$\|K_t^n - S_{t-s} K_s^n\|_{L^2(\Omega;H)} \le C_T |t-s|^{3/4}, \qquad 0 \le s < t \le T. \tag{2}$$

Proof. Let p_t be the Dirichlet kernel and $d(y) = y \wedge (1-y)$. For $0 < a \le 2T$, the reflection formula for the interval and comparison with the two half-lines at the nearer boundary give

$$p_a(z,y) \le C a^{-1/2} e^{-|z-y|^2/(Ca)}, \qquad p_a(y,y) \asymp a^{-1/2}\big(1 \wedge d(y)^2/a\big). \tag{3}$$

For example, if $d(y) \le \sqrt{a}$, the half-line expression $(2\pi a)^{-1/2}(1 - e^{-2d(y)^2/a})$ is comparable to $d(y)^2 a^{-3/2}$; if $d(y) \ge \sqrt{a}$, the reflected terms are exponentially smaller than the free diagonal term. The finitely many remaining images are absorbed in a constant depending only on T. Hence, for $\rho_\tau(y) = \int_0^\tau p_{2r}(y,y)\,dr$,

$$\rho_\tau(y) \asymp \sqrt{\tau} \wedge d(y), \qquad \int_0^1 \rho_\tau(y)^{-1/2} dy \le C_T \tau^{-1/4}. \tag{4}$$

Indeed the integral of $r^{-1/2}(1\wedge d^2/r)$ is comparable to $\sqrt{\tau}$ when $d\ge\sqrt{\tau}$, and to d otherwise. Also

$$\sup_z \int_0^1 p_a(z,y)\rho_\tau(y)^{-1/2}dy \le C_T(a^{-1/4}+\tau^{-1/4}). \tag{5}$$

To see this, use $\rho_\tau^{-1/2}\le C(\tau^{-1/4}+d^{-1/2})$. The first part integrates to $C\tau^{-1/4}$. For the second, bound $d(y)^{-1/2}\le y^{-1/2}+(1-y)^{-1/2}$ and use $y=\sqrt{a}\,r$ near the left boundary (similarly near the right) to get $Ca^{-1/4}\sup_{\alpha\ge 0}\int_0^\infty e^{-(r-\alpha)^2/C}r^{-1/2}dr<\infty$. Finally, for $0<\tau\le T$,

$$\|\rho_\tau^{-1}S_\tau f\|_2 \le C_T\tau^{-1/2}\|f\|_2. \tag{6}$$

Here $\rho_\tau^{-1}\le C(\tau^{-1/2}+d^{-1})$; $S_\tau f\in H_0^1$; Hardy's inequality gives $\|S_\tau f/d\|_2\le C\|\partial_x S_\tau f\|_2$; and $\|\partial_x S_\tau f\|_2\le C\tau^{-1/2}\|f\|_2$.

Write $D^n_{s,t}=K^n_t-S_{t-s}K^n_s$. Freeze the equation at time s:

$$A^n_{s,t}=\int_s^t S_{t-r}b_n(S_{r-s}u^n_s+V_r-S_{r-s}V_s)\;dr. \tag{7}$$

For fixed n, if Π is a partition of $[s,t]$, then

$$\sum_{[a,b]\in\Pi} S_{t-b}A^n_{a,b}\to D^n_{s,t} \quad \text{in } L^2(H). \tag{8}$$

Indeed the summand differs from $\int_a^b S_{b-r}b_n(u^n_r)\,dr$ by at most $C_n(b-a)^2$ in $L^2(H)$, because b_n is Lipschitz and the two arguments differ only by the drift accumulated on $[a,r]$, whose H-norm is at most $C_n(r-a)$. After applying S_{t-b} and summing, the error is $O_n((t-s)|\Pi|)$.

The estimates below are uniform in n. Given $\mathcal{F}_s$,

$$Z_r(y):=S_{r-s}u^n_s(y)+V_r(y)-S_{r-s}V_s(y)$$

is Gaussian with variance $\rho_{r-s}(y)$. Since b_n is a probability density, $\sup_m \mathbb{E}b_n(m+N(0,\sigma^2))\le C\sigma^{-1}$. Moreover, given $\mathcal{F}_r$, the future part of $Z_q(z)$, $q>r$, has variance

$$\int_r^q\int_0^1 p_{q-\ell}(z,\xi)^2\,d\xi d\ell=\int_0^{q-r}p_{2a}(z,z)\,da=\rho_{q-r}(z).$$

Thus, for $s<r<q$,

$$\mathbb{E}_s\, b_n(Z_r(y))b_n(Z_q(z))\le C\rho_{r-s}(y)^{-1/2}\rho_{q-r}(z)^{-1/2}. \tag{9}$$

Using $\int p_{t-r}(x,y)p_{t-q}(x,z)dx=p_{2t-r-q}(y,z)$, then (5) in y and (4) in z, we obtain

$$\mathbb{E}_s\|A^n_{s,t}\|_2^2\le C\int_s^t\int_r^t\big((2t-r-q)^{-1/4}+(r-s)^{-1/4}\big)(q-r)^{-1/4}\,dq\,dr\le C|t-s|^{3/2}. \tag{10}$$

The last inequality is just scaling on the triangle $0\le r-s\le q-s\le t-s$. Hence $\|A^n_{s,t}\|_{L^2H}\le C|t-s|^{3/4}$.

For $s < u < t$, let $\delta A^n_{s,u,t} = A^n_{s,t} - S_{t-u}A^n_{s,u} - A^n_{u,t}$. The pieces before u cancel. For $r > u$, the two frozen arguments differ by $S_{r-u}D^n_{s,u}$. Conditional on $\mathcal{F}_u$, the remaining noise at space point y has variance $\rho_{r-u}(y)$, and $\|(b_n * \varphi_\sigma)'\|_\infty \le C\sigma^{-2}$, uniformly in n. Therefore

$$|\mathbb{E}_u\{b_n(Z_r(y)) - b_n(Z_r(y) + h(y))\}| \le C\rho_{r-u}(y)^{-1}|h(y)|.$$

With $h = S_{r-u}D^n_{s,u}$, (6) and contraction of S_t imply

$$\|\mathbb{E}_u\delta A^n_{s,u,t}\|_2 \le C(t-u)^{1/2}\|D^n_{s,u}\|_2. \tag{11}$$

Taking $\mathbb{E}_s$ and using Jensen gives the same estimate with $\mathbb{E}_s$ in place of $\mathbb{E}_u$.

We need only the following dyadic sewing fact. Suppose $A_{s,t}$ is H-valued and $\mathcal{F}_t$-measurable, and for every dyadic interval $[a,b]$ with midpoint u,

$$\|\mathbb{E}_a\delta A_{a,u,b}\|_{L^2H} \le \Gamma_1(b-a)^{1+\varepsilon_1}, \quad \|\delta A_{a,u,b} - \mathbb{E}_a\delta A_{a,u,b}\|_{L^2H} \le \Gamma_2(b-a)^{1/2+\varepsilon_2}.$$

If $J_m = \sum_i S_{t-t_{i+1}}A_{t_i,t_{i+1}}$ over the 2^m equal subintervals of $[s,t]$, then J_m converges and

$$\|\mathcal{A}_{s,t} - A_{s,t}\|_{L^2H} \le C(\Gamma_1|t-s|^{1+\varepsilon_1} + \Gamma_2|t-s|^{1/2+\varepsilon_2}). \tag{12}$$

Indeed, $J_{m+1} - J_m$ is the sum of $-S_{t-b}\delta A_{a,u,b}$. The conditional-mean parts are bounded by the triangle inequality, giving a factor $2^m(|t-s|2^{-m})^{1+\varepsilon_1}$. The centered parts are orthogonal: for two level-m intervals with the first lying earlier, the earlier summand is $\mathcal{F}_a$-measurable for the later one, while the later summand has conditional mean zero given $\mathcal{F}_a$. Thus the square norm is the sum of the square norms. Summing the resulting geometric series proves (12).

For fixed n, $Y_n(h) = \sup_{0<t-s\le h}\|D^n_{s,t}\|_{L^2H}/|t-s|^{3/4}$ is finite, because b_n is bounded. On intervals of length at most h, (11) gives

$$\|\mathbb{E}_a\delta A^n_{a,u,b}\|_{L^2H} \le CY_n(h)(b-a)^{5/4}.$$

Also

$$\|\delta A^n_{a,u,b} - \mathbb{E}_a\delta A^n_{a,u,b}\|_{L^2H} \le 2\|\delta A^n_{a,u,b}\|_{L^2H} \le C(b-a)^{3/4},$$

because δA is a sum of three A-terms and (10) applies to each. The dyadic sewing limit is $D^n_{s,t}$ by (8). Combining (10) and (12),

$$\|D^n_{s,t}\|_{L^2H} \le C|t-s|^{3/4} + CY_n(h)|t-s|^{5/4}.$$

Thus $Y_n(h) \le C + Ch^{1/2}Y_n(h)$. Choose $h_0 > 0$, depending only on T, so that $Ch_0^{1/2} \le 1/2$; then $Y_n(h_0) \le 2C$, uniformly in n. For longer intervals use the additivity $D^n_{s,t} = S_{t-r}D^n_{s,r} + D^n_{r,t}$ and split into pieces of length at most h_0. This gives a linear bound $C_T|t-s|$, which is bounded by $C_T|t-s|^{3/4}$ on $0 \le t-s \le T$. The lemma follows. $\square$

The reversible reference measure and its gap

Let γ be Brownian bridge measure on E, the centered Gaussian measure with covariance $(-\Delta_D)^{-1}$. Define

$$\Lambda(v) = \int_0^1 \mathbf{1}_{\{v(r)>0\}}\, dr, \qquad \pi(dv) = Z^{-1} e^{2\Lambda(v)}\, \gamma(dv). \tag{13}$$

For $B_n' = b_n$, $B_n(a) = \int_{-\infty}^a b_n(r)dr$, put

$$\pi_n(dv) = Z_n^{-1} \exp\left(2\int_0^1 B_n(v(r))\, dr\right) \gamma(dv).$$

Since $0 \le B_n \le 1$, $B_n(a) \to \mathbf{1}_{\{a>0\}}$ for $a \neq 0$, and a Brownian bridge spends zero Lebesgue time at a fixed level,

$$\|\pi_n - \pi\|_{\mathrm{TV}} \to 0. \tag{14}$$

All densities $d\pi_n/d\gamma$ and $d\pi/d\gamma$ are bounded above and below by deterministic positive constants.

Lemma 2. *There are constants $C, c > 0$, independent of n, such that for every bounded continuous f, g and every $t \ge 0$,*

$$\int P_t^n f\, d\pi_n = \int f\, d\pi_n, \qquad \int g P_t^n f\, d\pi_n = \int f P_t^n g\, d\pi_n, \tag{15}$$

and

$$\mathrm{Var}_{\pi_n}(P_t^n f) \le e^{-ct}\, \mathrm{Var}_{\pi_n}(f). \tag{16}$$

Proof. Let Π_m be projection onto the first m Dirichlet modes and set

$$U_{n,m}(v) = \int_0^1 B_n(\Pi_m v(r))\, dr.$$

Its H-gradient is $\nabla_H U_{n,m}(v) = \Pi_m b_n(\Pi_m v)$. Let $P_t^{n,m}$ be the semigroup of

$$u_t^{n,m} = S_t x + V_t + \int_0^t S_{t-r}\Pi_m b_n(\Pi_m u_r^{n,m})\, dr.$$

The first m Fourier coordinates solve the finite-dimensional Langevin diffusion

$$dX_j = -(\lambda_j/2) X_j\, dt + \partial_j U_{n,m}(X)\, dt + d\beta_j, \qquad j \le m,$$

and the tail is an independent OU process. Hence, by finite-dimensional integration by parts and the standard OU integration by parts in the tail, $P_t^{n,m}$ is reversible with respect to $\pi_{n,m} = Z_{n,m}^{-1} e^{2U_{n,m}} \gamma$, and its Dirichlet form is $\mathcal{E}_{n,m}(F,F) = \frac{1}{2}\int \|\nabla_H F\|_2^2\, d\pi_{n,m}$ on smooth cylinder functions.

The measures $\pi_{n,m}$ converge to π_n in total variation: $\Pi_m v \to v$ in H, γ-a.s., and B_n is bounded and Lipschitz. For fixed n, $u_t^{n,m} \to u_t^n$ in probability in E. In H, this follows from Gronwall, since $\Pi_m b_n(\Pi_m z)$ is uniformly Lipschitz in z for fixed n, and for each H-valued y, $\Pi_m b_n(\Pi_m y) \to b_n(y)$ in H, dominated by $2\|b_n\|_\infty$. For convergence in E at a fixed $t > 0$, split the drift convolution into $[0, t-\varepsilon]$ and $[t-\varepsilon, t]$. On the first part, $S_{t-r} : H \to H_0^\alpha \subset E$, $\alpha > 1/2$, is bounded. On the short terminal part,

$$\int_{t-\varepsilon}^{t} \|S_{t-r}F_r\|_\infty dr \leq C\|b_n\|_\infty \int_0^\varepsilon a^{-1/4} da,$$

for both $F_r = \Pi_m b_n(\Pi_m u_r^{n,m})$ and $F_r = b_n(u_r^n)$, because Π_m is an L^2-contraction and $S_a : H \to E$ has norm $Ca^{-1/4}$. This tends to zero with ε, uniformly in m.

Consequently the reversible identities pass to $m = \infty$. For instance, if $h_m = d\pi_{n,m}/d\gamma$, then $h_m \to h_n = d\pi_n/d\gamma$ in $L^1(\gamma)$ and the h_m's are uniformly bounded; together with $P_t^{n,m} f(x) \to P_t^n f(x)$ and bounded convergence this gives (15).

It remains to record the gap uniformly. The Gaussian Poincare inequality for γ in H-gradient form is

$$\mathrm{Var}_\gamma(F) \leq \lambda_1^{-1} \int \|\nabla_H F\|_2^2 \, d\gamma$$

on smooth cylinder functions. Since all densities $d\pi_{n,m}/d\gamma$ lie in a fixed interval $[a_0, a_0^{-1}]$, the Holley–Stroock bounded perturbation argument gives

$$\mathrm{Var}_{\pi_{n,m}}(F) \leq C\, \mathcal{E}_{n,m}(F, F),$$

with C independent of n, m. For a symmetric Markov semigroup this Poincare inequality is equivalent to exponential L^2-decay, so

$$\mathrm{Var}_{\pi_{n,m}}(P_t^{n,m} f) \leq e^{-ct}\, \mathrm{Var}_{\pi_{n,m}}(f).$$

Letting $m \to \infty$, using the convergence just proved and the uniformly bounded densities, yields (16). $\square$

Proposition 1. *The measure π is P_t-invariant, P_t is symmetric on $L^2(\pi)$, and*

$$\mathrm{Var}_\pi(P_t f) \leq e^{-ct}\, \mathrm{Var}_\pi(f), \qquad f \in L^2(\pi),\ t \geq 0. \tag{17}$$

Proof. For $f, g \in C_b(E)$, (1), (14), (15), and the uniform boundedness of the densities imply, by dominated convergence with respect to γ,

$$\int P_t f \, d\pi = \lim_n \int P_t^n f \, d\pi_n = \int f \, d\pi, \qquad \int g P_t f \, d\pi = \int f P_t g \, d\pi.$$

On the Polish space E, these identities identify the measure $\int \pi(dx) P_t(x, \cdot)$ with π, and the measure $\pi(dx)P_t(x, dy)$ with its transpose; hence invariance and detailed balance hold for all bounded Borel functions. Similarly, (16) passes to the limit for bounded continuous f. Finally $C_b(E)$ is dense in $L^2(\pi)$, and an invariant Markov operator is an L^2-contraction, so symmetry and (17) extend to all $L^2(\pi)$. $\square$

Fixed-time absolute continuity

Lemma 3. *For every deterministic $x \in E$ and every $t > 0$,*

$$P_t(x, \cdot) \ll \gamma,$$

and hence $P_t(x, \cdot) \ll \pi$.

Proof. Let $\nu_t^x = \mathrm{Law}(S_t x + V_t)$. We first bound $\mathrm{Ent}(P_t^n(x, \cdot) \mid \nu_t^x)$ uniformly in n. Put $r_k = t(1 - 2^{-k})$, $\Delta_k = r_{k+1} - r_k$, and $D_k^n = K_{r_{k+1}}^n - S_{\Delta_k} K_{r_k}^n$. Define

$$F_s^n = \sum_{k \geq 0} \frac{\mathbf{1}_{(r_{k+1}, r_{k+2}]}(s)}{\Delta_{k+1}} S_{s - r_{k+1}} D_k^n. \tag{18}$$

The process is predictable because D_k^n is $\mathcal{F}_{r_{k+1}}$-measurable. Lemma 1 and contraction of S_t give

$$\mathbb{E} \int_0^t \|F_s^n\|_2^2 ds \leq C_t \sum_{k \geq 0} \frac{\Delta_k^{3/2}}{\Delta_{k+1}} < \infty, \tag{19}$$

uniformly in n. For partial sums,

$$\int_0^t S_{t-s} F_s^{n,N} ds = \sum_{k=0}^{N} S_{t - r_{k+1}} D_k^n = S_{t - r_{N+1}} K_{r_{N+1}}^n,$$

where $F^{n,N}$ denotes the sum in (18) up to $k = N$. Since $r_{N+1} \uparrow t$, the regularized mild solution gives K^n continuous with values in E, and $S_a \to I$ strongly on E,

$$\int_0^t S_{t-s} F_s^n \, ds = K_t^n \tag{20}$$

in probability in E (and hence in H). The state space assertion is legitimate because, for every deterministic $F \in L^2(0, t; H)$,

$$\sum_j \lambda_j^\alpha \left| \int_0^t e^{-\lambda_j (t-s)/2} F_j(s) ds \right|^2 \leq \sum_j \lambda_j^{\alpha - 1} \int_0^t |F_j(s)|^2 ds \leq C \|F\|_{L^2(0,t;H)}^2$$

for $0 < \alpha < 1$; choosing $\alpha > 1/2$ gives an element of $H_0^\alpha(0,1) \subset E$.

We use the following standard entropy form of Girsanov, and recall its proof. If F is predictable, H-valued, and has bounded energy, then

$$X^F = S_t x + V_t + \int_0^t S_{t-s} F_s ds$$

satisfies

$$\mathrm{Ent}(\mathrm{Law}(X^F) \mid \nu_t^x) \leq \frac{1}{2} \mathbb{E} \int_0^t \|F_s\|_2^2 ds. \tag{21}$$

For the first M Brownian coordinates this is finite-dimensional Girsanov: under the tilted measure, the shifted coordinates $\beta_j + \int_0^\cdot F_j(s)ds$, $j \le M$, are Brownian, and the entropy cost is $\frac{1}{2}\mathbb{E}\int_0^t \sum_{j\le M} F_j(s)^2 ds$. Applying data processing to the deterministic-kernel stochastic convolution gives the same bound for the first M terminal Fourier coordinates. Since the Borel sigma-field of E is generated by these coordinates and relative entropy is the supremum over increasing finite-coordinate sigma-fields, (21) follows.

Localize F^n by $\tau_R = \inf\{s : \int_0^s \|F_r^n\|_2^2 dr > R\}$ and $\widehat{F}^{n,R} = F^n \mathbf{1}_{[0,\tau_R]}$. Let

$$X_R = S_t x + V_t + \int_0^t S_{t-s}\widehat{F}_s^{n,R} ds.$$

Then (21) and (19) give

$$\mathrm{Ent}(\mathrm{Law}(X_R) \mid \nu_t^x) \le \frac{1}{2}\,\mathbb{E}\int_0^t \|F_s^n\|_2^2 ds. \tag{22}$$

On $\{\tau_R > t\}$, (20) gives $X_R = u_t^n$, while $\mathbb{P}(\tau_R \le t) \le R^{-1}\mathbb{E}\int_0^t \|F_s^n\|_2^2 ds \to 0$. Hence $X_R \to u_t^n$ in probability in E. Lower semicontinuity of relative entropy on the Polish space E, then (19) and (22), yield

$$\sup_n \mathrm{Ent}(P_t^n(x,\cdot) \mid \nu_t^x) < \infty. \tag{23}$$

By the assumed weak convergence (1) and lower semicontinuity again, $\mathrm{Ent}(P_t(x,\cdot) \mid \nu_t^x) < \infty$. Thus $P_t(x,\cdot) \ll \nu_t^x$.

It remains only to compare Gaussian measures. In the basis $e_j(r) = \sqrt{2}\sin(j\pi r)$, $\lambda_j = j^2\pi^2$, the covariance of ν_t^x has eigenvalues

$$q_j(t) = \frac{1 - e^{-\lambda_j t}}{\lambda_j},$$

whereas γ has eigenvalues λ_j^{-1}. The ratios $1 - e^{-\lambda_j t}$ are bounded away from zero and differ from 1 by a square-summable sequence. The Cameron–Martin spaces are therefore both $H_0^1(0,1)$, and $S_t x \in H_0^1$. Feldman–Hajek gives $\nu_t^x \sim \gamma$. $\square$

Conclusion

Let η be an invariant probability measure. If $\pi(A) = 0$, then Lemma 3 gives $P_t(y, A) = 0$ for every $y \in E$, and hence

$$\eta(A) = \int_E P_t(y, A)\,\eta(dy) = 0.$$

Thus $\eta \ll \pi$; write $d\eta = h\,d\pi$. Detailed balance gives the L^1-duality

$$\int f\,P_t g\,d\pi = \int P_t f\,g\,d\pi$$

for bounded Borel f and $g \in L^1(\pi)$, by truncating g. Therefore

$$\int fh\,d\pi = \int P_t f\,d\eta = \int P_t f\,h\,d\pi = \int f\,P_t h\,d\pi,$$

so $P_t h = h$ in $L^1(\pi)$. For $a > 0$, let $h_a = h \wedge a$. Since $r \mapsto r \wedge a$ is concave,

$$P_t h_a \le (P_t h) \wedge a = h_a.$$

The two sides have the same π-integral; therefore $P_t h_a = h_a$. Now $h_a \in L^2(\pi)$, and the spectral gap (17) implies $\mathrm{Var}_\pi(h_a) = 0$. Thus h_a is constant for every a. Letting $a \uparrow \infty$ and using $\int h\,d\pi = 1$, we obtain $h = 1$ π-a.s. Hence $\eta = \pi$.

Therefore P_t has at most one invariant probability measure on $C_0([0,1])$. In fact, if an invariant probability exists, it is the Gibbs measure π in (13).

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